Database Management Lab

(ENCS205)

Bachelor of Technology

in

Computer Science and Engineering

Submitted by

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PRACTICAL 3

DDL Questions (Data Definition Language)

1. Create the TEACHER table with the following attributes: (Name, DeptNo, Date_of_Joining, DeptName, Location, Salary).
2. Create the SALES table with the following attributes: (Sales_No, Salesname, Branch, Salesamount, DOB).
3. Create the DEPARTMENTS table with the following attributes: (dept_no, dept_name, dept_location).
4. Create the EMPLOYEES table with the following attributes: (emp_id, emp_name, emp_salary, dept_no).

```sql
CREATE DATABASE LAB3;
USE LAB3;

CREATE TABLE TEACHER (
    Name VARCHAR(50),
    DeptNo INT,
    Date_of_Joining DATE,
    DeptName VARCHAR(50),
    Location VARCHAR(50),
    Salary DECIMAL(10,2)
);

CREATE TABLE SALES (
    Sales_No INT PRIMARY KEY,
    Salesname VARCHAR(50),
    Branch VARCHAR(50),
    Salesamount DECIMAL(10,2),
    DOB DATE
);

CREATE TABLE DEPARTMENTS (
    dept_no INT PRIMARY KEY,
    dept_name VARCHAR(50),
    dept_location VARCHAR(50)
);

CREATE TABLE EMPLOYEES (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(50),
    emp_salary DECIMAL(10,2),
    dept_no INT,
    FOREIGN KEY (dept_no) REFERENCES DEPARTMENTS(dept_no)
);
```

DML Questions (Data Manipulation Language)

From TEACHER Table

1. Insert five records into the **TEACHER** table.
2. Give an increment of 25% salary to teachers working in the Mathematics department.
3. Give an increment of 15% salary to teachers working in the Commerce department.

```sql
INSERT INTO TEACHER VALUES
('Anita', 1, '2015-06-12', 'Mathematics', 'Delhi', 50000),
('Rajesh', 2, '2017-03-20', 'Commerce', 'Mumbai', 45000),
('Sneha', 1, '2018-07-15', 'Mathematics', 'Delhi', 52000),
('Amit', 3, '2016-01-10', 'Science', 'Chennai', 48000),
('Priya', 2, '2019-09-05', 'Commerce', 'Mumbai', 47000);
SET SQL_SAFE_UPDATES = 0;
UPDATE TEACHER
SET Salary = Salary + (Salary * 0.25)
WHERE DeptName = 'Mathematics';

UPDATE TEACHER
SET Salary = Salary + (Salary * 0.15)
WHERE DeptName = 'Commerce';
```

From SALES Table

1. Insert five records into the **SALES** table.

```sql
INSERT INTO SALES VALUES
(101, 'Rohan', 'Delhi', 20000, '1995-12-21'),
(102, 'Meena', 'Mumbai', 35000, '1993-05-15'),
(103, 'Aakash', 'Delhi', 15000, '1992-12-09'),
(104, 'Pooja', 'Chennai', 40000, '1994-08-18'),
(105, 'Karan', 'Mumbai', 30000, '1991-12-11');
```

2. Calculate the total sales amount in each branch.

```sql
SELECT Branch, AVG(Salesamount) AS Average_Sales
FROM SALES
GROUP BY Branch;
```

Branch	Average_Sales
Delhi	17500.000000
Mumbai	32500.000000
Chennai	40000.000000

3. Calculate the average sales amount in each branch.

```
SELECT Branch, AVG(Salesamount) AS Average_Sales
FROM SALES
GROUP BY Branch;
```

Branch	Average_Sales
Delhi	17500.000000
Mumbai	32500.000000
Chennai	40000.000000

4. Display the Salesname and DOB of salesmen born in the month of December with date format like *21-Dec-09*.

```
SELECT Salesname,
DATE_FORMAT(DOB, '%d-%b-%y') AS Formatted_DOB
FROM SALES
WHERE MONTH(DOB) = 12;
```

Salesname	Formatted_DOB
Rohan	21-Dec-95
Aakash	09-Dec-92
Karan	11-Dec-91

5. Display the Salesname and DOB of salesmen in alphabetical order of the month.

```
SELECT Salesname, DOB
FROM SALES
ORDER BY MONTHNAME(DOB);
```

Salesname	DOB
Pooja	1994-08-18
Rohan	1995-12-21
Aakash	1992-12-09
Karan	1991-12-11
Meena	1993-05-15

From JOIN Operations

1. Display employee details and department details where the department numbers are the same in both tables.

```
SELECT *
FROM EMPLOYEES E
INNER JOIN DEPARTMENTS D
ON E.dept_no = D.dept_no;
```

emp_id	emp_name	emp_salary	dept_no	dept_no	dept_name	dept_location
201	Arjun	60000.00	1	1	Mathematics	Delhi
204	Simran	50000.00	1	1	Mathematics	Delhi
202	Neha	55000.00	2	2	Commerce	Mumbai
203	Vikas	45000.00	3	3	Science	Chennai

2. Display the employee name and department name using a LEFT OUTER JOIN.

```
SELECT E.emp_name, D.dept_name
FROM EMPLOYEES E
LEFT JOIN DEPARTMENTS D
ON E.dept_no = D.dept_no;
```

emp_name	dept_name
Arjun	Mathematics
Neha	Commerce
Vikas	Science
Simran	Mathematics

3. Display the employee name and department name using a RIGHT OUTER JOIN.

```
SELECT E.emp_name, D.dept_name
FROM EMPLOYEES E
RIGHT JOIN DEPARTMENTS D
ON E.dept_no = D.dept_no;
```

emp_name	dept_name
Arjun	Mathematics
Simran	Mathematics
Neha	Commerce
Vikas	Science

4. Display the details of employees who draw a salary greater than the average salary.

```
SELECT *
FROM EMPLOYEES
WHERE emp_salary > (SELECT AVG(emp_salary) FROM EMPLOYEES);
```

emp_id	emp_name	emp_salary	dept_no
201	Arjun	60000.00	1
202	Neha	55000.00	2
NULL	NULL	NULL	NULL

TCL Questions (Transaction Control Language)

1. Perform the **ROLLBACK** command on the TEACHER table.
2. Perform the **COMMIT** command on the TEACHER table.
3. Write a query to implement a **SAVEPOINT**.

```
ROLLBACK;
COMMIT;
SAVEPOINT sp1;
```

DCL Questions (Data Control Language)

1. Write a query to **grant all privileges** on the EMPLOYEES table to DEPARTMENTS.

```
DROP USER 'DEPARTMENTS'@'localhost';
CREATE USER 'DEPARTMENTS'@'localhost' IDENTIFIED BY 'pass123';
```

2. Write a query to **grant selected privileges** on the EMPLOYEES table to DEPARTMENTS.

```
REVOKE ALL PRIVILEGES
ON LAB3.EMPLOYEES
FROM 'DEPARTMENTS'@'localhost';

GRANT SELECT, INSERT
ON LAB3.EMPLOYEES
TO 'DEPARTMENTS'@'localhost';
```

3. Write a query to **revoke all privileges** on the EMPLOYEES table from DEPARTMENTS.

```
REVOKE ALL PRIVILEGES
ON LAB3.EMPLOYEES
FROM 'DEPARTMENTS'@'localhost';
```

4. Write a query to **revoke selected privileges** on the EMPLOYEES table from DEPARTMENTS.

```
GRANT SELECT, INSERT
ON LAB3.EMPLOYEES
TO 'DEPARTMENTS'@'localhost';
REVOKE SELECT, INSERT
ON LAB3.EMPLOYEES
FROM 'DEPARTMENTS'@'localhost';
FLUSH PRIVILEGES;
```